

1 INTRODUCTION

1.1 STUDY OBJECTIVES

- To define, describe, and forecast the global microbiome sequencing services market based on service, type, technology, end user, and region
- To provide detailed information regarding the major factors influencing the market growth (such as drivers, restraints, challenges, and opportunities)
- To strategically analyze micromarkets¹ with respect to individual growth trends, prospects, and contributions to the overall microbiome sequencing services market
- To analyze the opportunities in the market for stakeholders and provide details of the competitive landscape for market leaders operating in the microbiome sequencing services market
- To forecast the revenue of the market with respect to five regional segments—North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa
- To profile the key market players and comprehensively analyze their service portfolios, market positions, and core competencies²
- To track and analyze competitive developments such as services launches, expansions, agreements, collaborations, partnerships, and acquisitions in the microbiome sequencing services market

1.2 MARKET DEFINITION

Microbiome sequencing is applied to decode the genetic material of all the microorganisms in a given sample which may contain bacteria, viruses, or fungi. Whole-genome sequencing (WGS) and amplicon sequencing are the two key microbiome sequencing services. Sequencing-based detection of microorganisms exhibits different advantages over traditional approaches, some of which are mentioned below:

- Detection of anaerobic bacteria
- More cost-effective microbiome characterization of multiplex microbial samples
- Identification of non-cultivable bacteria
- Estimate the relative abundance of microbes in a sample
- Profile hundreds of microorganisms in a single analysis
- Examine complex microbiomes and microbiota communities

The microbiome sequencing services market comprises services employed for sample preparation, sequencing, library preparation, and data analysis suitable for different samples, including human microbiome samples and environmental samples.

1. Micromarkets are the further segments and subsegments of the microbiome sequencing services market.

2. Core competencies of companies are captured in terms of the key developments, market shares, and key strategies adopted to sustain their positions in the market.

1.2.1 INCLUSIONS AND EXCLUSIONS

PARTICULAR	INCLUSION	EXCLUSION
Service	Qualitative and quantitative analysis (in terms of USD million) of microbiome sequencing services, including sample preparation, sequencing and library preparation, and data analysis.	NA
Type	Qualitative and quantitative analysis (in terms of USD million) of microbiome sequencing service types, such as amplicon sequencing, whole-genome sequencing, shotgun sequencing, and de novo sequencing.	NA
Technology	Qualitative and quantitative analysis (in terms of USD million) of technologies, such as sequencing by synthesis (SBS), sequencing by ligation (SBL), nanopore sequencing, and semiconductor sequencing.	NA
End User	Qualitative and quantitative analysis (in terms of USD million) of microbiome sequencing service end users such as pharmaceutical and biotechnology companies, academic and research institutes, small-scale CROs, and CDMOs.	NA

1.3 MARKET SCOPE

1.3.1 MARKETS COVERED

MICROBIOME SEQUENCING SERVICES MARKET			
 SERVICE - Sample Preparation - Sequencing and Library Preparation - Data Analysis	 TYPE - Amplicon Sequencing - Whole-genome Sequencing - Shotgun Sequencing - Other Types	 TECHNOLOGY - Sequencing by Synthesis - Sequencing by Ligation - Nanopore Sequencing - Other Technologies	
 END USER - Pharmaceutical and Biotechnology Companies - Academic and Research Institutes - Other End Users	 REGION - North America (US and Canada) - Europe (Germany, UK, France, Italy, Spain, and Rest of Europe) - Asia Pacific (Japan, China, India, and Rest of Asia Pacific) - Latin America (Brazil and Rest of Latin America) - Middle East and Africa		

Note 1: Other types of microbiome sequencing services include de novo sequencing (whole-genome sequencing without any reference genome), metagenomic sequencing, and metatranscriptomic sequencing.

Note 2: Other microbiome sequencing technologies include semiconductor sequencing and Sanger sequencing.

Note 3: Other end users include hospitals and clinics, small-scale CROs, and CDMOs.

Note 4: The Rest of Europe includes Switzerland, Greece, the Netherlands, Belgium, Denmark, and Ireland.

Note 5: The Rest of Asia Pacific includes South Korea, Australia, and countries in Southeast Asia.

Note 6: The Rest of Latin America includes Argentina, Peru, Chile, and Colombia.